

Ecom Express Dashboard

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1. Project Overview

This project focuses on analyzing the sales performance of **Ecom Express**, an e-commerce retailer, by developing an interactive business intelligence dashboard using **Power BI**. The dashboard consolidates sales, customer, product, and order data into a single, intuitive interface, enabling stakeholders to monitor key business metrics and evaluate overall performance. It provides a comprehensive view of revenue generation, average order value, order cancellations, product-wise and category-wise sales, geographic revenue distribution, and quarterly sales trends. Interactive filters for **product category** and **purchase date** allow users to explore the data dynamically, uncover patterns, compare performance across different segments, and identify areas requiring attention. By transforming raw transactional data into meaningful visual insights, the dashboard supports data-driven decision-making, helping management optimize sales strategies, improve operational efficiency, and enhance overall business performance.

2. Business Problem

E-commerce businesses generate a massive volume of transactional data every day, including customer purchases, product sales, and order fulfillment records. As operations expand across multiple product categories and geographical regions, manually tracking business performance becomes increasingly challenging. Without a centralized reporting solution, decision-makers often struggle to obtain a clear and timely understanding of key business metrics, making it difficult to identify trends, evaluate operational efficiency, and respond proactively to emerging issues.

Some of the major business challenges include:

- Monitoring overall revenue, order volume, and average order value.
- Identifying the highest-performing products and product categories.
- Measuring the financial impact of cancelled orders.
- Comparing sales performance across different states.
- Analyzing quarterly revenue trends to identify seasonal patterns.
- Enabling data-driven strategic planning through real-time performance monitoring.

To address these challenges, this project leverages **Power BI** to transform raw sales, customer, and product data into an interactive business intelligence dashboard. The dashboard consolidates critical performance indicators into a single interface, allowing users to explore data through dynamic filters and visualizations. By providing actionable insights into sales performance, cancellations, product trends, and regional revenue distribution, the solution empowers stakeholders to make informed decisions, optimize business strategies, and improve overall operational efficiency.

Dataset Description

The dashboard is built using three datasets.

Customers

Contains customer information including:

- Customer ID
- Customer Name
- State
- Mobile Brand
- Operating System

Orders

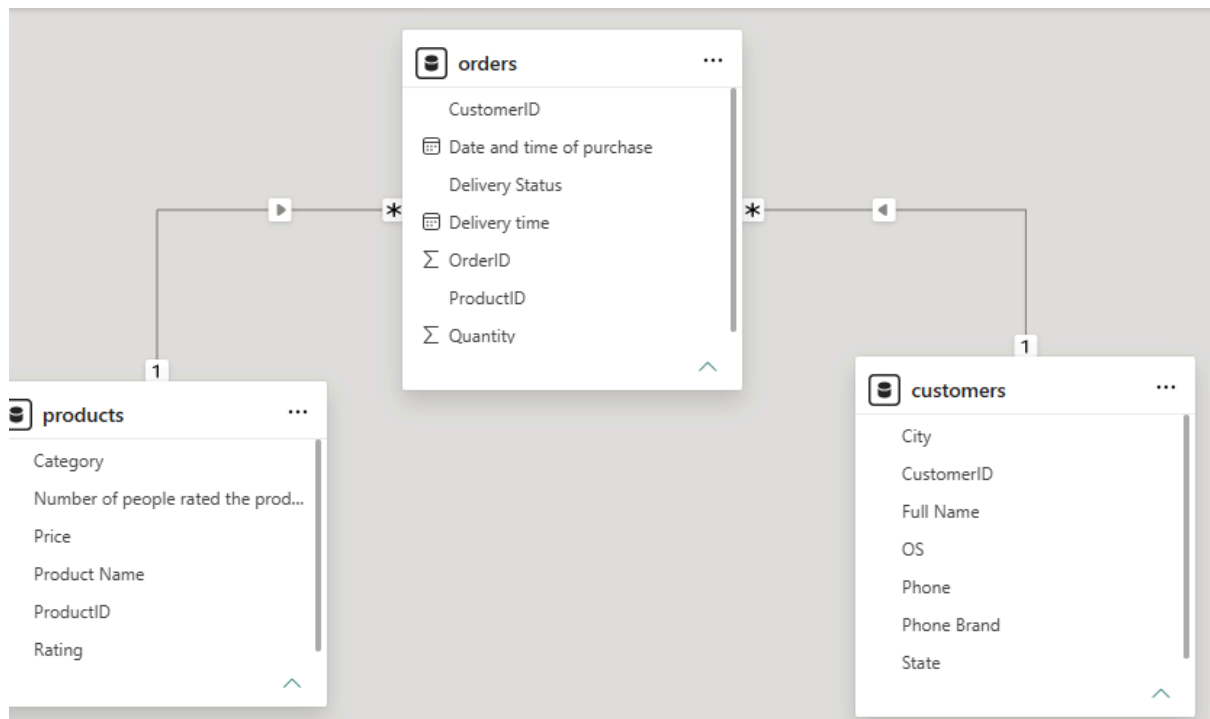
Contains transaction details such as:

- Order ID
- Customer ID
- Product ID
- Purchase Date
- Quantity
- Revenue

Products

Contains product-related information including:

- Product Name
- Category
- Price
- Ratings



Data Cleaning & Transformation

Data cleaning was performed in **Power Query** before building the dashboard.

Customers Table

The following transformations were performed:

- Removed the unnecessary last column.
- Created a custom column named **Operating System**:
 - Apple devices were classified as **iOS**.
 - All remaining mobile brands were classified as **Android**.
- Standardized all state names by converting them to **Proper Case**.
- Merged **First Name** and **Last Name** into a single **Customer Name** column.

These transformations improved data consistency and simplified reporting.

Dashboard KPIs

The dashboard tracks the following Key Performance Indicators:

KPI	Value
Revenue	\$1.25 Billion
Average Order Value	\$112.85K
Cancellation Rate	29.72%
Revenue Lost Due to Cancellations	\$525.42 Million
Total Orders	15.69K

Dashboard Features

The dashboard includes:

- KPI Cards for overall business performance.
- Category slicer for interactive filtering.
- Purchase date filter.
- Revenue by Product Name.
- Revenue by Product Category.
- Revenue by State.
- Quarterly Revenue Trend.



Business Insights

Revenue Overview

The company generated approximately **\$1.25 billion** in total revenue from **15.69K orders**, indicating a strong sales volume.

High Cancellation Rate

Nearly **30% of all orders were cancelled**, resulting in an estimated revenue loss of **\$525.42 million**.

Business Recommendation

- Investigate reasons for order cancellations.
- Improve inventory availability.
- Strengthen delivery and payment processes.
- Offer proactive customer support to reduce cancellations.

Best Performing Product Categories

The **Laptop** category generated the highest revenue among all categories.

Other strong-performing categories include:

- Mobile
- Headphones
- Tablet

This suggests that premium electronics contribute significantly to overall sales.

Top Revenue-Generating Products

The highest revenue was generated by products such as:

- MacBook
- OnePlus 9
- Sony Headphones
- HP Pavilion
- Samsung devices

These products represent key revenue drivers and should receive focused marketing and inventory planning.

Geographic Sales Analysis

The states contributing the highest revenue include:

- Maharashtra
- Gujarat
- Rajasthan
- West Bengal
- Tamil Nadu

These regions represent the strongest markets and can be prioritized for targeted marketing campaigns and promotional offers.

Quarterly Revenue Trend

Revenue remained relatively stable during **Quarter 1**, **Quarter 2**, and **Quarter 3**, but experienced a noticeable decline in **Quarter 4**.

Possible reasons include:

- Seasonal demand fluctuations
- Increased cancellations
- Lower customer purchasing activity

Further investigation into Quarter 4 sales patterns can help identify opportunities to improve year-end performance.

Business Recommendations

Based on the dashboard analysis, the following recommendations can improve business performance:

- Reduce order cancellations by improving inventory management and delivery reliability.
- Focus marketing efforts on high-performing states.
- Maintain adequate stock levels for top-selling products.
- Expand product offerings within high-revenue categories such as Laptops and Mobiles.
- Analyze Quarter 4 sales trends to understand the causes of declining revenue.
- Use customer purchasing trends to design targeted promotional campaigns.

Tools & Technologies

The dashboard was developed using **Power BI Desktop** as the primary business intelligence platform. **Power Query** was utilized to clean, transform, and prepare the raw data for analysis, while **DAX (Data Analysis Expressions)** was used to create calculated measures and key performance indicators (KPIs). A well-structured **data model** was designed to establish relationships between multiple datasets, and **CSV files** served as the primary data source for the project.

- Power BI Desktop
- Power Query
- DAX (Data Analysis Expressions)
- Data Modeling
- CSV Files

Skills Demonstrated

This project demonstrates the complete business intelligence workflow, starting from data preparation to dashboard development. It involved cleaning and transforming raw data, building an optimized data model, creating DAX measures for KPI calculations, and designing an interactive dashboard that enables users to explore business performance effectively. The dashboard provides meaningful visualizations and actionable insights to support data-driven decision-making.

- Data Cleaning
- Data Transformation
- Power Query
- Data Modeling
- DAX Measures
- KPI Design
- Interactive Dashboard Development
- Business Intelligence
- Data Visualization
- Business Insight Generation